POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 1011 | Semester I | Practicum | 4.5 Credits

Fundamentals of Civil Engineering

A construction-focused handbook covering building planning, masonry, foundations, roofing, plumbing, domestic electrical services and fire safety—supported by field activities, drawings and workshop procedures.

Programmes	**Civil streams**
Teaching scheme	**3:0:3:0**
Contact hours	**90**
Theory / Practical	**45 / 39 listed**
CIE / ESE	**40 / 100**
Official source	**3-page PDF**

Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied SITTTR Kerala Diploma Curriculum Revision 2026 PDF for Course 1011. No Revision 2021 lesson content has been merged.

Verified identity

- **Title:** Fundamentals of Civil Engineering
- **Programmes:** Civil Engineering, Civil Engineering & Planning, Civil & Rural Engineering, Civil & Environmental Engineering, Civil Engineering (Construction Technology)
- **Semester:** I
- **Type:** Practicum
- **Prerequisite:** NIL

Official-source inconsistencies / omissions

- The course table states **90 contact hours**, while module allocations total **45 lecture + 39 practical = 84 hours**. The missing 6 hours are not assigned to a module in the supplied PDF.
- The assessment summary is consistent at **40 CIE + 100 ESE = 140**, split as Theory 20+60 and Practical 20+40.
- The detailed syllabus stops during Module II on page 3; detailed learning-outcome rows for the remaining Module II topics and Modules III–IV are not present in the supplied file.
- Official textbooks, references, equipment list, explicit safety specification and model question paper: **Not specified in the supplied official syllabus PDF.**

Display decision: The website uses Semester I, 4.5 credits and 90 total contact hours because these are stated in the course-description table. Module cards retain the exact listed module hours and visibly disclose the 6-hour mismatch.

Course Dashboard

4.5

Credits

90

Total contact hours

3 + 3

Weekly L + P

140

Total assessment marks

Teaching and assessment scheme

L	T	P	S	Theory CIE	Theory ESE	Practical CIE	Practical ESE	Total
3	0	3	0	20	60	20	40	140

Compact jump menu

How to Use This Handbook

1. Understand
Read the concept and definitions.

2. Visualise
Study the drawing, table or process.

3. Practise
Follow calculations and workshop steps.

4. Verify
Answer self-check and expected questions.

Malayalam support: □□□ topic-□□□ □□□□□□ English technical explanation □□□□□□□□□□. □□□□□ diagram □□□□□□□ Malayalam support □□□□□□□□□□□ concept □□□□□□□□□□□□. Practical □□□□□□□□□□□□ instructor-approved safety procedure □□□□□□□□ □□□□□□□□□□□□.

Objectives and Course Outcomes

Official objectives

1. To impart basic principles of building construction, planning, and regulations.

2. To familiarize the masonry and finishing works used in building construction.
3. To familiarize the various components of buildings.
4. To gain practical knowledge of various building services.

Official Course Outcomes

1. **CO1:** Apply fundamental civil engineering principles in surveying, area measurement, regulations and building planning.
2. **CO2:** Practice appropriate masonry and finishing techniques for durable and quality building construction.
3. **CO3:** Apply knowledge of building components, foundations, and roofing materials in various construction scenarios.
4. **CO4:** Demonstrate basic building service concepts related to plumbing, electrical and fire safety in buildings.

Bloom's taxonomy level for all four outcomes: **Apply**.

Curriculum Coverage Map

Module	Official topics	Hours L/P	CO	Cognitive emphasis	Handbook section
I	Civil engineering relevance; buildings; site selection; areas, KBR concepts; surveying	11/9	CO1	Remember, Understand, Apply	Planning and surveying
II	Brick/stone masonry, bonds, pointing, plastering, painting	12/12	CO2	Remember, Understand, Apply	Masonry and finishes
III	Building components, shallow/deep foundations, roofing materials	12/9	CO3	Apply	Components and foundations
IV	Plumbing, sanitary systems, domestic electricity, wiring and fire safety	10/9	CO4	Apply	Services and fire safety

Learning roadmap

Site + rules + survey

Masonry + finishes

Components + foundations + roof

Plumbing + electrical + fire safety

MODULE I · CO1

Fundamentals of Civil Engineering

11 L 9 P

Civil engineering converts social needs into safe, serviceable and durable infrastructure. This module begins with the profession, then moves to building classification, site selection, regulatory area terms and basic surveying.

1. Relevance and major disciplines

Civil engineering supports shelter, transportation, water supply, sanitation, irrigation, energy infrastructure and environmental protection. Major disciplines include structural, geotechnical, transportation, water resources, environmental, construction management and surveying/geomatics.

Application: A housing project requires structural design, soil investigation, road access, drainage, quantity estimation and site supervision. No single discipline works in isolation.

Malayalam support: Civil Engineering □□□□□□□ building □□□□□□□ □□□□. Road, bridge, water supply, drainage, soil, environment, construction management □□□□□□□□□□□□□□ □□□□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□□ field □□□.

2. Buildings by occupancy and construction type

Occupancy classification groups buildings according to their principal use. Typical NBC groups include residential, educational, institutional, assembly, business, mercantile, industrial, storage and hazardous. Always verify the current approved NBC/Kerala rule classification before statutory submission.

Load-bearing structure

Walls carry floor and roof loads to the foundation. Suitable for limited spans and low-rise construction. Openings and later alterations require care.

Framed structure

Beams and columns carry loads; walls mainly enclose spaces. Offers larger openings and flexible planning but needs engineered detailing.

Malayalam support: Load-bearing structure-□ wall □□□□□ load foundation-□□□□□□□ □□□□□□□□□□□. Framed structure-□ beam-column frame □□□ □□□□□□□ load path; wall □□□□□□□ enclosure □□□.

3. Site selection

- Confirm legal title, access and permitted land use.
- Study ground slope, drainage, flood risk and soil condition.
- Check water, electricity, road, sewer/septic feasibility and emergency access.
- Consider orientation, ventilation, sunlight, noise and future expansion.
- Avoid unstable slopes, filled lowland and uncontrolled watercourses unless engineered mitigation is approved.

Common mistake: Choosing a cheap plot before checking access width, flood level and soil bearing condition can make the project uneconomical.

4. Building area concepts, Coverage and FSI

Plinth area

Covered built-up area measured at floor level, generally including wall thickness according to the applicable measurement rule.

Built-up area

Total constructed floor area; exact inclusions must follow the applicable authority's definitions.

Floor area

Usable covered area measured within the building after prescribed exclusions.

Carpet area

Net usable area where a carpet can be laid, excluding wall thickness and specified service/common areas.

Coverage (%) = (Ground-floor covered area ÷ Plot area) × 100
FSI = Total floor area on all floors ÷ Plot area

Solved example

Given: Plot = 300 m², ground coverage = 120 m², total floor area = 240 m².

Coverage = 120/300 × 100 = **40%**.
FSI = 240/300 = **0.80**.

These calculations do not prove statutory compliance; compare with the current Kerala Building Rules and sanctioning authority limits.

Malayalam support: Coverage ground level-□ plot □□□□ □□□□□□ building cover □□□□□□□□□□ □□□□□□□□□. FSI □□□□□ floor area-□□□□□□□□ □□□□□□ plot area-□□□□ □□□□□□□□ □□□□□□□□□□□□□□.

5. Surveying principles and classification

Surveying determines the relative positions of points on, above or below the earth's surface. Two basic principles are: work from whole to part, and fix a point by at least two independent measurements.

Basis	Examples
Instrument	Chain/tape, compass, plane table, levelling, theodolite, total station, GNSS
Function	Topographical, cadastral, engineering, route, hydrographic
Nature of field	Land, marine, astronomical
Earth model	Plane surveying for limited areas; geodetic surveying includes earth curvature

Common units: millimetre, metre, kilometre; square metre, hectare; degree-minute-second or decimal degree.

Practical: classify nearby buildings

1. Select a safe observation route and obtain permission.
2. Record building use, apparent construction system and number of storeys.
3. Classify occupancy using the approved class list supplied by the instructor.
4. Tabulate evidence; do not enter private property without permission.

Practical: tape distance

1. Inspect tape, ranging rods and ground condition.
2. Range the line between intervisible stations.
3. Keep tape aligned, horizontal where required and under consistent pull.
4. Mark full tape lengths, measure remainder, repeat and compare.

> Distance = (Number of full tape lengths × tape length) + remainder

MODULE II · CO2

Masonry and Finishes

12 L 12 P

Masonry quality depends on bond, line, level, plumb, jointing, material condition and curing—not merely on placing units one above another.

1. Brick masonry terms and requirements

Important terms include stretcher, header, bed, face, back, arris, frog, course, bed joint, perpend, closer, quoin and lap. Good masonry uses sound units, appropriate mortar, full joints, proper bond, controlled joint thickness, accurate line/level/plumb and adequate curing.

> **Common defects:** continuous vertical joints, dry bricks absorbing mortar water, thick irregular joints, unfilled joints, poor corner lead and inadequate curing.

> **Malayalam support:** Bond □□□□□□□ brick pattern □□□□□□□ □□□□; vertical joint □□□□□□□□□□□□ □□□□□□□□□□□□□□ load distribute □□□□□□□□□□□ wall-□□□□ stability □□□□□□□□□□□ □□□□□□□□□□ arrangement □□□.

2. Header, stretcher, English and Flemish bonds

Bond	Arrangement	Typical note
Stretcher	All visible faces are stretchers	Common for half-brick partitions
Header	All visible faces are headers	Useful where wall thickness suits header length
English	Alternate header and stretcher courses	Strong and comparatively easy to control
Flemish	Alternate headers and stretchers in each course	Attractive face; requires skilled workmanship

Drawing note: Show closers, lap and corner arrangement exactly as instructed in the workshop drawing. A decorative brick pattern without correct bonding is not acceptable.

3. Stone masonry

Rubble masonry uses roughly dressed or undressed stones with comparatively irregular joints. **Ashlar masonry** uses accurately dressed rectangular blocks with thin, regular joints. Selection depends on appearance, available stone, skill, cost and structural requirement.

4. Pointing, plastering and painting

Pointing finishes exposed masonry joints to improve weather resistance and appearance. Types in the syllabus: flush, recessed, weathered, keyed/grooved.

Plastering provides a smooth, protective and sanitary surface. A good plaster should adhere, remain even, resist cracking and suit the exposure.

Paint protects and decorates surfaces. Typical ingredients are base, pigment, vehicle/binder, solvent/thinner, drier and additives. An ideal paint should spread well, adhere, dry reasonably, retain colour and resist the intended environment.

> **Malayalam support:** Pointing masonry joint □□□□□□□ finish □□□□□□□□□□□. Plastering □□□□□□□ surface cover □□□□□□□□□□□. Painting-□ surface preparation □□□□□□□□□ □□□□ paint □□□□□□□□□□□□□□□ peeling □□□□□□□□□.

Brickwork practical controls

- Use instructor-approved mortar and brick soaking method.
- Build corner leads first; check gauge, line, level and plumb frequently.
- Maintain bond and prescribed lap; fill bed and vertical joints fully.
- Keep work height uniform and cure as instructed.

Enamel and emulsion painting

1. Identify substrate and isolate the work area.
2. Clean, repair and dry the surface; remove loose material.
3. Apply compatible primer/undercoat where required.
4. Stir paint, apply thin even coats, respect drying time and ventilation.
5. Inspect coverage, runs, brush marks and adhesion.

> Use PPE and follow the product safety data sheet. Do not use flames or sparks near solvent-based coatings.

MODULE III · CO3

Introduction to Building Components

12 L | 9 P

1. Substructure and superstructure

The **substructure** lies mainly below ground/plinth level and transfers loads safely to soil. The **superstructure** is the portion above plinth level.

> **Malayalam support:** Foundation soil-□□□□□□□ load □□□□□□□□□□□. Plinth ground level-□□□ floor level-□□□ □□□□□□□ transition □□□. Lintel opening-□□□□ □□□□□□□□ load support □□□□□□□□□□□□; sunshade □□□□□□ □□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□.

2. Foundations and their purpose

A foundation distributes structural loads within allowable soil pressure, limits total and differential settlement, provides stability against sliding/overturning and creates a level base.

Type	Basic use / idea
Wall footing	Continuous support below a load-bearing wall
Isolated column footing	One footing under one column
Combined footing	One footing supports two or more columns
Strap footing	Two footings linked by a strap beam to balance eccentricity
Grillage foundation	Layers of beams spread heavy concentrated load
Raft foundation	Large slab supporting many columns/walls over most of plan area
Pile foundation	Deep elements transfer load by end bearing, shaft resistance or both
Well foundation	Deep foundation commonly associated with bridge piers and water works

Foundation choice cannot be made from appearance or a simple rule of thumb. It requires structural loads, soil investigation, groundwater and settlement assessment by qualified personnel.

3. Roofing materials

Material	Advantages	Limitations / checks
RCC	Durable, rigid, usable terrace	Dead load, waterproofing, thermal gain
Mangalore Pattern (MP) tiles	Ventilated, traditional appearance	Requires supporting framework; breakage
Ceramic roofing tiles	Durable finish, thermal comfort	Detailing, pitch and support spacing
Thatched	Low embodied energy, thermal comfort	Fire, decay, maintenance and local restrictions
GI / corrugated GI sheets	Light, quick installation	Heat, noise, corrosion, wind fastening
Plastic / fibre sheets	Light and daylight options	UV ageing, fire performance, fixing compatibility

Innovative materials should be compared using weight, span, durability, fire performance, thermal comfort, acoustic performance, maintenance, cost and recyclability.

Drawing and report activities

- Prepare a neatly labelled building component sketch.
- Draw plan and section of isolated, combined and raft foundations using the scale and conventions specified by the instructor.
- Survey innovative roofing products; record manufacturer, material, unit weight, recommended slope, support spacing, warranty, fire classification and local availability. Verify data from current product documents rather than guessing.

MODULE IV · CO4

Building Services and Fire Safety

10 L 9 P

1. Plumbing and sanitary systems

A plumbing system conveys water to fixtures and removes wastewater safely. Components include pipes, joints, valves, taps, traps, fixtures, supports, inspection points and testing arrangements. Sanitary systems require water seals, ventilation and gradients appropriate to the approved design.

Component	Purpose
Pipe	Conveys water, waste or vent air
Fitting	Changes direction, size or connection
Fixture	User interface such as wash basin, sink, WC
Trap	Maintains water seal against foul gases
Valve / tap	Controls flow

Malayalam support: Pipe flow □□□□□□□□□□□□□□□□; fitting direction/connection □□□□□□□□□□□; fixture user □□□□□□□□□□□□□□ unit □□□. Trap-□□□□□ water seal foul gas room-□□□□□□ □□□□□□□□ □□□□□□□□.

2. Domestic electrical supply basics

Phase

Live conductor carrying alternating voltage relative to neutral/earth.

Neutral

Return/reference conductor connected to the supply neutral system.

Earth

Protective conductor intended to carry fault current and keep exposed metal near earth potential.

Load consumes electrical power. A **circuit** is a complete conductive path. Single-phase supply is common for ordinary domestic loads; three-phase supply is used for larger loads and motors. Exact voltage, earthing and connection requirements must follow the utility and current electrical rules.

3. Protection devices and wiring systems

- **Rewirable fuse:** melts a replaceable element under excessive current; correct rating and approved replacement are essential.
- **Cartridge fuse:** enclosed fuse element with controlled characteristics.
- **MCB:** automatically opens on overload/short circuit and can be reset after the fault is cleared.
- **Earth leakage circuit breaker / residual-current protection:** disconnects when leakage imbalance exceeds its operating threshold; it does not replace overcurrent protection.

System	Basic idea	Typical context
Cleat wiring	Insulated conductors supported on cleats	Temporary installations only where permitted
Batten wiring	Cables fixed on battens with clips	Simple surface wiring
Casing and capping	Conductors enclosed in casing with removable cover	Surface wiring; material/rule dependent
Conduit wiring	Conductors drawn through rigid/flexible conduit	Common concealed or surface protection

Wire selection depends on current capacity, voltage drop, insulation, ambient temperature, grouping, installation method, fault protection and mechanical conditions. It must be designed by competent persons.

Electrical safety: Students must not work on an energised installation. Isolate, lock out where applicable, verify absence of voltage and work only under authorised supervision.

4. Fire safety requirements

Fire safety combines prevention, detection, warning, safe escape, containment and fire-fighting access. Core checks include unobstructed exits, suitable travel paths, emergency lighting/signage, alarm arrangements, extinguishers appropriate to the hazard, electrical housekeeping, safe storage, fire-resisting separation and regular drills.

> **Malayalam support:** Fire extinguisher □□□□□□□ fire safety □□□□. ignition prevent □□□□□□□□, alarm □□□□□□, □□□□□□□ safely evacuate □□□□□□□□, fire spread limit □□□□□□□□, fire service access □□□□□□□□□□□□□—□□ □□□□□□□ system □□□ □□□□.

Practical: pipe and tap fitting

1. Identify pipe material, size, fittings and approved jointing method.
2. Measure and mark; cut square using the correct tool.
3. Deburr and clean; prepare threads or solvent joint as instructed.
4. Assemble without cross-threading or excessive force.
5. Support the line and perform an instructor-approved leak test.

Practical: GI/PVC thread cutting

Threading methods differ by material. GI typically uses dies and cutting lubricant; many PVC systems use solvent-welded fittings and should not be threaded unless the specific pipe/fitting system permits it. Follow the workshop manual and product instructions.

> Secure work in a proper vice, wear eye protection, keep hands clear of dies/cutters and remove sharp burrs.

College fire-safety audit report

1. Record building name, date, team and permission.
2. Map exits, stairs, assembly area and fire-service access.
3. Check alarm points, emergency lights, signs and extinguisher locations.
4. Note obstruction, expiry/service labels and housekeeping issues without operating equipment.
5. Attach photographs only with permission; classify findings as compliant, observation or urgent risk.
6. Recommend corrective action and responsible follow-up.

Procedure / Rule Bank

Area calculations

Draw a clear boundary sketch, divide irregular shapes, retain units, state inclusions/exclusions and verify totals.

Masonry control

Material → mortar → bond → line → level → plumb → joints → curing.

Foundation decision

Loads + soil + groundwater + settlement + constructability + durability.

Roof comparison

Weight, slope, span, water tightness, heat, noise, fire, corrosion, maintenance.

Plumbing quality

Correct material, alignment, slope, joint, support, trap seal, access and testing.

Electrical / fire

Isolate; protect against overcurrent and leakage; maintain escape routes and alarms.

Practical / Activity Centre

Activity	Record headings	Key quality/safety check
Building classification survey	Aim, location, observations, occupancy, structure type, conclusion	Permission and no unsafe access
Area computation	Sketch, dimensions, calculations, result	Units and boundary assumptions
Tape measurement	Stations, tape length, runs, remainder, repeated value	Alignment, slope and consistent pull
Brick bond jobs	Bond sketch, materials, procedure, checks, result	Line, level, plumb, lap, joints, curing
Painting	Surface, coating system, coats, drying, defects	Ventilation, PPE, ignition control
Foundation drawings	Scale, plan, section, labels, dimensions	Drawing convention and legibility
Pipe/tap fitting	Material, tools, joint, test, result	Correct joining method and leak test
Fire-safety report	Scope, checklist, findings, evidence, actions	Do not operate life-safety equipment

Generic record format

Aim:

Tools / materials:

Safety precautions:

Procedure:

Observation / calculation:

Result and inference:

Instructor verification:

Expected / Practice Questions with Answers

Module I

Differentiate Coverage and FSI.

Coverage compares ground-floor covered area with plot area and is expressed as a percentage. FSI compares total floor area of all floors with plot area and is a ratio.

State two principles of surveying.

Work from whole to part; locate a point by at least two independent measurements.

Why is site drainage important?

It limits flooding, dampness, erosion, foundation problems and access failure.

Module II

Differentiate English and Flemish bond.

English bond alternates complete header and stretcher courses. Flemish bond alternates headers and stretchers within each course.

Why are masonry joints fully filled?

To transfer load, improve weather resistance, reduce voids and create uniform support.

Pointing versus plastering?

Pointing finishes exposed joints; plastering covers the complete surface.

Module III

State four purposes of a foundation.

Spread load, limit settlement, provide stability, and provide a firm level base.

When is raft foundation considered?

When individual footings would occupy a large area, soil pressure is low, or settlement control requires a common base—subject to engineering design.

List criteria for roofing material comparison.

Weight, durability, slope, fire performance, thermal/acoustic comfort, maintenance, cost and availability.

Module IV

What is the purpose of protective earth?

To provide a low-impedance fault-current path and reduce dangerous touch voltage on exposed conductive parts.

Does an ELCB/RCD replace an MCB?

No. Leakage protection and overcurrent protection address different hazards; coordinated devices are required.

Name the main elements of fire safety.

Prevention, detection, alarm, escape, containment, suppression and emergency access.

Application questions

1. A 250 m² plot has 100 m² ground coverage and 225 m² total floor area. Calculate Coverage and FSI. **Answer:** 40%; 0.90.
2. A wall shows continuous vertical joints. Identify the defect and corrective principle. **Answer:** Poor bond; provide proper overlap/lap according to the selected bond.
3. A wash basin smells despite cleaning. Which plumbing component should be checked first? **Answer:** Trap water seal and venting.
4. An MCB repeatedly trips. Should it be replaced by a higher rating? **Answer:** No. Isolate and investigate overload, short circuit, cable sizing and appliance faults through a competent electrician.

Practice Model Paper — not an official SITTTR model question paper

The supplied PDF does not specify an official model-paper pattern. Use these as mixed practice, not as an official marks distribution.

1. Explain the relevance and major disciplines of civil engineering.
2. Classify buildings by occupancy and construction system.
3. Calculate Coverage and FSI from given dimensions.
4. Explain English and Flemish bonds with sketches.
5. Compare rubble and ashlar masonry.
6. Describe the functions of building components.
7. Compare shallow and deep foundations.
8. Prepare a roofing-material comparison table.

9. Draw the domestic electrical supply block diagram and state the purpose of protection devices.
10. Prepare a fire-safety inspection checklist for a college building.

Quick Revision

M1

Profession → building classification → site → area rules → surveying.

M2

Units + mortar + bond + joints + finishing + curing.

M3

Components → load path → foundation → roof selection.

M4

Water and waste → electrical supply/protection → fire safety.

Exam and practical checklist

- Write units in every area and distance calculation.
- Use labelled sketches rather than unsupported descriptions.
- Separate official facts from engineering judgement.
- Never recommend a foundation or wire size without required design data.
- In practical records, include safety, observation and result—not only procedure.

Glossary

Term	Meaning
Arris	Sharp edge formed by two surfaces of a brick or stone.
Course	Horizontal layer of masonry units.
Quoin	External corner of masonry.
Plinth	Portion between surrounding ground and finished floor level.
Lintel	Horizontal member over an opening.
FSI	Ratio of total floor area to plot area.
Trap	Plumbing fitting retaining water seal.
MCB	Miniature Circuit Breaker for overload/short-circuit protection.
Earth	Protective connection to the earthing system.

References

Official references: Not specified in the supplied official syllabus PDF.

Supplementary study direction: Use the current National Building Code, Kerala Building Rules, approved Indian Standards, utility rules, manufacturer installation instructions and institution workshop/laboratory manuals under instructor guidance. These are supplementary and are not presented here as official syllabus references.